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# HandDGCL: Two-hand 3D reconstruction based disturbing graph contrastive learning

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## Abstract

Virtual reality (VR) and augmented reality (AR) applications are becoming increasingly prevalent. However, constructing realistic 3D hands, especially when two hands are interacting, from a single RGB image remains a major challenge due to severe mutual occlusion and the enormous diversity of hand poses. In this article, we propose a disturbing graph contrastive learning strategy for two-hand 3D reconstruction. This involves a graph disturbance network designed to generate graph feature pairs to enhance the consistency of the two-hand pose features. A contrastive learning module leverages high-quality generative features for a strong feature expression. We further propose a similarity distinguish method to divide positive and negative features for accelerating the model convergence. Additionally, a multi-term loss is designed to balance the relation among the hand pose, the visual scale and the viewpoint position. Our model has achieved state-of-the-art results in the InterHand2.6M benchmark. Ablation studies show the model's great ability to correct unreasonable hand movements. In subjective assessments, our graph disturbance learning method significantly improves the construction of realistic 3D hands, especially when two hands are interacting.

**KEYWORDS**

graph contrastive learning, hand pose estimation, hand shape reconstruction

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## 1 | INTRODUCTION

With the development of virtual reality and augmented reality (VR/AR), hand poses are widely used for virtual-real interaction in many scenarios. Recently, hand shape reconstruction, which is considered an extension of hand pose estimation, has drawn a lot of attention due to its realistic demonstration. Previous studies[1,2] have successfully employed 3D hand techniques to visualize hand joints. However, visualizing two interacting hands still presents significant challenges (Figure 1).

Many existing studies have concentrated on single-hand reconstruction from RGB,[1-4] depth,[5] or sparse keypoints.[6,7] When dealing with interacting two-hand reconstruction, these single hand-based methods have poor performance, since it increases the difficulty including inter-hand collisions and mutual occlusions. Recently, some large-scale interacting hand datasets are released to support the two-hand shape reconstruction. Two-hand-shape-pose[8] and IHMR[9] reconstruct two-hands by estimating MANO[10] parameters, which are later mapped to triangular hand meshes using a pre-defined statistical model (i.e., MANO). IntagHand[11] directly regresses a fixed number of mesh vertex coordinates using a graph convolutional network (GCN). Nevertheless, few effective methods are proposed to resolve the problem of two-hand 3D reconstruction, as the complex occlusion and deformation occur from the interaction of two hands.

In this article, we present a graph contrastive learning based approach to reconstruct two interacting hands, where an explicit graph disturbing strategy is performed at the feature level to augment the hand feature. The explicit disturbances change the external properties of the hand graph, such as graph dimension, graph edge and some feature noise. To effectively utilize these features, we incorporate a graph contrastive learning for generating strong hand-feature expression. We also propose a similarity distinguish approach to divide positive and negative features for accelerating the model convergence. Finally, a new loss function is designed to enhance 3D visual quality by balancing relation among the hand pose, the visual scale, and the viewpoint position. Experimental results prove that our approach achieves a qualitative improvement compared to existing methods on subjective and objective evaluation.

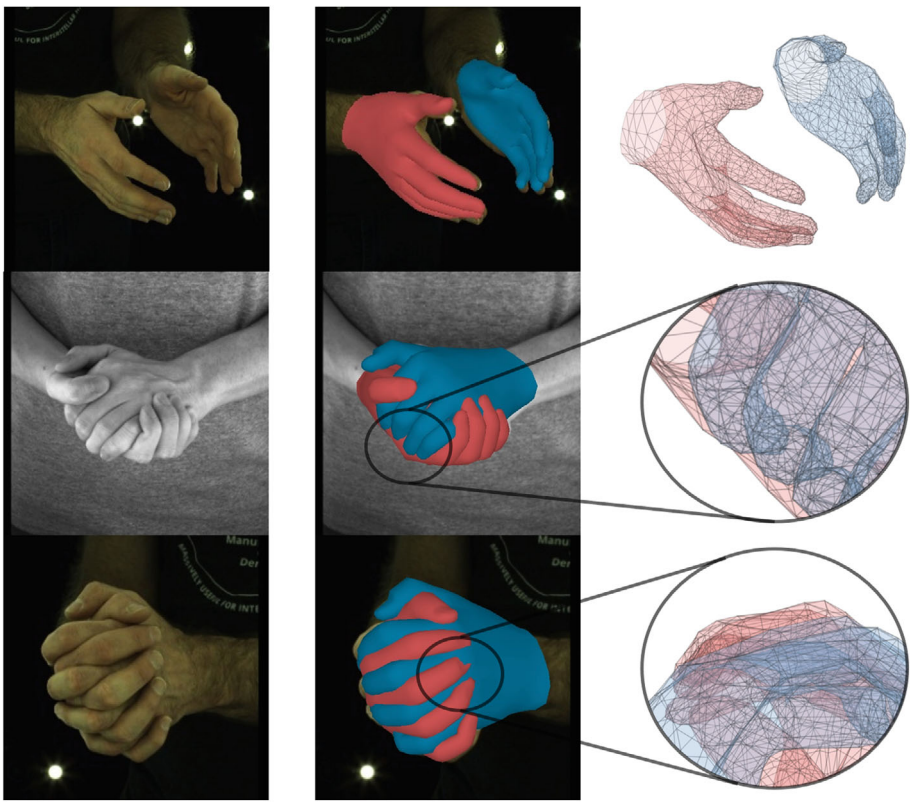


**FIGURE 1** Interactive and non-interactive hand shape reconstruction from an RGB image. Geometric deformation occurs in case of the two-hand interaction, which limits visual quality of 3D hand reconstruction.

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Overall, our contributions are summarized as follows:

- We propose a novel graph disturbance learning to solve the problem of 3D two hands reconstruction by generating graph feature pairs without compromising consistency to hand pose. Explicit as well as implicit graph disturbance is added, which improves the expression of the features.
- We incorporate a scheme of similarity control in order to smooth the learning curve and reduce divergence within the contrastive learning structure.
- We design a multi-term loss function that deals with the pose, the scale and the camera parameter separately so that the model can balance the weight of each task.

## 2 | RELATED WORKS

According to different downstream task, 3D hand field can be categorized into three: (1) 3D hand pose estimation; (2) 3D hand shape reconstruction, (3) hand-object interactions. There are numerous datasets[12-18] proposed these years of above three, which greatly boost the machine understanding of human hands.

### 2.1 | 3D hand pose estimation

Previous works mainly focus on the depth domain[19-22] and the RGB domain.[1,3,15,22-25] Compared with 2D hand, 3D hand pose has more complexity and lack one dimension of information for single RGB naturally. Recently, 3D hand pose estimation from a monocular RGB image has achieved great progress.[18] To alleviate the confusion of uniform appearance in hands, the method in Reference 26 uses ResGCN based refinement and conditional adversarial learning scheme to fully exploit hand features. Reference 27 associates keypoints in the heatmap with hand joints using multi-head self-attention to predict complex hand interactions. The work of Reference 28 shows that contrastive training scheme extracts better hand feature representation especially for those vague or easily confused cases.

### 2.2 | Graph CNN based hand shape reconstruction

GCN[29] has been proved that it is stable to convey 3D hand shape information with less computational expense. Reference 2 proposed a pipeline to make mesh generation from coarse to precise via GCN in hand shape reconstruction. In Reference 30, RGB images are encoded into embeddings of a graph morphable model, which helps the reconstruction of 3D hand from RGB space. These two works mentioned above are the first to construct 3D hand mesh by GCN, and both stand for the great ability of GCN in hand shape reconstruction. After them, GCN is widely used in this aera. Reference 31 designs two transformer-based modules to predict the shape of hands at the occlusion part. Reference 11 adds cross-hand attention modules behind the GCN operation within three blocks to refine the representation of two-hand interactions.

### 2.3 | Contrastive learning

Self-supervised methods have been intensively studied in recent years. As one of them, contrastive learning is competitive due to its simple structure as well as high performance. Reference 32 makes it possible to catch up supervised methods in image classification using a simple framework of contrastive learning. The work of Reference 33 shows that pretraining with contrastive learning can also boost the performance of fine-tuning on a few labeled dataset. Reference 34 builds a dynamic dictionary with a queue and a moving-averaged encoder to enable a large and consistent dictionary that facilitates contrastive unsupervised representations learning. While these works rely on hard negative samples, Reference 35 cast contrastive pairs only using positive samples via a momentum encoder trained by the parameter update strategy of moving average, which makes pretraining easier and more friendly to low memory devices. These classical contrastive learning methods above all focus on the encoding of original data to extract better representation while

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our point is on the decoder. Before this article, Reference 36 builds hand representation by making expressive positive pairs on multiview images. They take the images of one hand in multiviews as positive samples that share the identical inherent representation, and learn these representations by the similar method of Reference 34. However, the geometric property of hands is not fully extracted and the contrastive pretraining cannot get regular performance when deals with complex hand poses or two-hand interactions. To solve these problems, we introduce graph contrastive learning to the field of 3D hand pose estimation and shape reconstruction in this article. As for graph contrastive learning, existing methods[37-41] excavate inherent features on graph in contrastive manners that inspire us utilize it in hand graph domain. We use a simple but powerful framework named SimSiam[42] that combines the advantages of classical contrastive learning, and mixes augmentations in hand graph domain to learn geometry-structured hand representation deeply.

## 3 | METHODOLOGY

### 3.1 | Overview

An overall structure of our model is illustrated in Figure 2. The main pipeline at the top of the figure shows the main workflow of the hand shape reconstruction task, which can be summarized as image feature extraction, graph feature conversion, disturbing graph contrastive learning and MANO-hand construction. The disturbing graph contrastive learning network (DGCLNet) module utilizes graph contrastive learning to generate pairs with consistent hand pose information, which is specially designed for graph-level visual contrastive learning task. Note that DGCLNet is only used to train the GCN, which means that DGCLNet are removed except the inside GCN during evaluation.

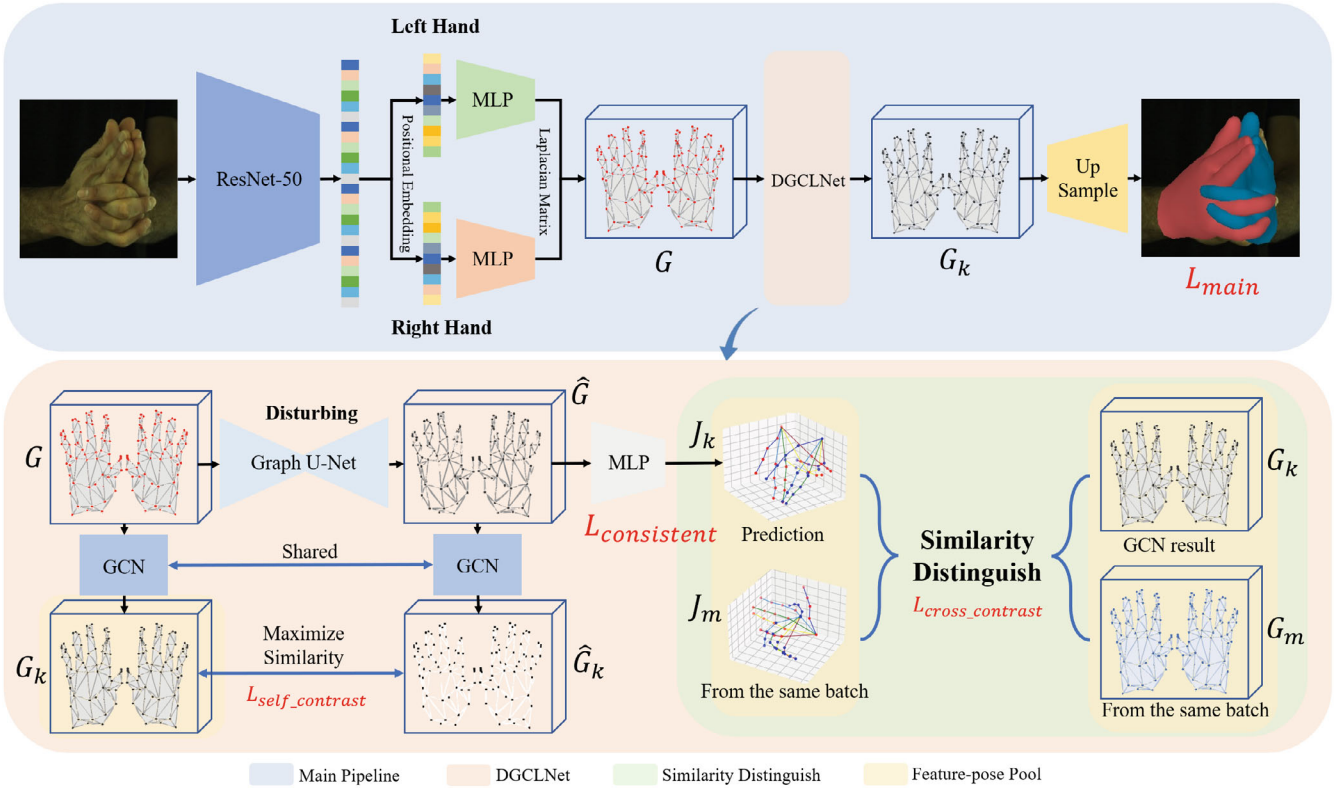



**FIGURE 2** Architecture overview. Given an RGB image, our model uses ResNet-50 for feature extraction and then the feature is divided into left hand and right hand via positional embedding. Both of the hand feature are further processed by two independent MLP layers, and build up the graph feature using the knowledge of Laplacian matrix. Next our DGCLNet makes disturbance by regenerating the graph feature through a graph u-net.[43] The pair of graph feature is passed through the same GCN, and the similarity between them is maximized. Finally the graph feature of original branch is thrown into the up sample layer to get the position of 778 vertices in standard MANO[10] way.

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The similarity distinguish is an extended module of DGCLNet to calculate similarity of a pair precisely from a feature-pose pool, and makes the model converge faster and better. Detailed information of the pipeline will be discussed in this section.

## 3.2 | Disturbing graph contrastive learning

Our model tackles the challenging visual task of two-hand shape reconstruction in a simply constructed end-to-end framework. The graph contrastive learning modules are proved to well decode image embeddings to hand mesh in the form of graph representation.

### 3.2.1 | Explicit graph disturbing strategy

Naturally, we consider using basic image-to-image data augmentation methods to perturb the target image features. In the location of the graph u-net in DGCLNet depicted in Figure 2, we applied the following four types of disturbance operations to the model. (a) Random mask on feature dimension. (b) Random noise on feature space. (c) Add random edges on graph. (d) Flip the edges on graph. After disturbance is added, the shared GCN processes the feature pair and a loss function is used to maximize the similarity of them.

However, there is an essential problem on steps above that is whether the intrinsic essence is changed by disturbance. It is well known that classical contrastive method does not change the intrinsic essence because the disturbance is mainly pixel-level and the effect of the disturbance makes slightly different on feature level. Therefore, the feature vector extracted by encoder maintain its consistency and contrastive method works out. But now we have the graph features distorted which makes it possible that the consistency of different branches is broken.

### 3.2.2 | Disturbing graph contrastive learning network

To avoid this problem, we design a new contrastive structure to generate pairs of disturbance as Figure 2 described, which corresponds to the $L_{\text{self_contrast}}$. The core of DGCLNet is a generative neural network essentially, which can be noted as $F_{\text{gen}}$. And we use graph u-net[43] in this article. The input of DGCLNet $\mathbf{G}$ is the original graph features which built by the previous network, and $\hat{\mathbf{G}}$ is the graph features after disturbance. All $\mathbf{G}$ are normalized before use, but for the sake of brevity, this step is omitted in the formula. Simply, we have

$$\hat{G} = F_{\text{gen}}(G), \tag{1}$$

The two graph features $\mathbf{G}$,$\hat{\mathbf{G}}$ are considered as a contrastive pair just like other contrastive learning method. $F_{\text{gen}}$ replaces those four kinds of simple graph-level operations above and the other setting remains the same. Therefore, the model feeds the contrastive pairs to the main GCN decoder and get graph features with lower feature dimension noted as $\mathbf{G_k}$,$\hat{\mathbf{G}}_\mathbf{k}$.

$$G_k = GCN(G), \hat{G}_k = GCN(\hat{G}), \tag{2}$$

$\mathbf{G_k}$,$\hat{\mathbf{G}}_\mathbf{k}$ are actually in a pool of features and poses, which accounts for their subscript. The similarity can be predicted as

$$S = G_k \cdot \hat{G}_k^{\top}, \tag{3}$$

The $\mathbf{S}$ here describes the similarity between one graph feature and its generative copy. $\mathbf{G_k}$ is passed to next part of the model while all other features in DGCLNet are abandoned.

But how to train DGCLNet? DGCLNet is also trained in end-to-end manner. In order to preserve pose information during training, we innovatively add a linear MLP after $\hat{\mathbf{G}}$ to predict hand pose labels which is taken as a subtask of hand reconstruction.

$$J_k = MLP(\hat{G}). \tag{4}$$

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It is because of the subtask that $\mathbf{G}$,$\hat{\mathbf{G}}$ are forced to maintain sufficient supervised information while $\mathbf{G}$ cannot be identical with $\hat{\mathbf{G}}$ due to the bottleneck structure of DGCLNet. Then we get a disturbance reproduction without losing the intrinsic essence of the hand pose naturally, which ensures the consistency that the contrastive learning scheme needs. Besides, the feature output by the graph u-net will not be identical with the input because of the bottleneck structure. Therefore this operation can be considered as disturbing. Note that the DGCLNet is only used during training, our goal is to get a great GCN decoder. The simplicity of the model during inference is also one of our advantages.

## 3.3 | Similarity distinguish

Based on the pre-estimate relative hand joints location, we intend to extend to the contrastive space of positive and negative samples, which can improve the cross-pose contrastive learning performance. According to the general contrastive learning paradigm, the ground truth of similarity between a positive sample pair is set to 1, while for a negative sample pair it is set to 0. To express it in a neat form, assuming similarity between a random pair from a feature-pose pool is noted by $\mathbf{S_p}$. The pool is defined as a dictionary of extracted features and their corresponding 3D poses as[34] did.

$$S_p[m,n] = G_m \cdot G_n^{\top}, \tag{5}$$

And we need it equals to 1 when the two of the pair are positive, equals to 0 when negative, which means the quantity of the similarity is discrete. However, we can make it continuous by defining a ground truth of similarity, which is a real number between 0 and 1. The pre-estimate relative hand joints location, noted $\mathbf{J}$, can be used to build the ground truth of similarity. Suppose we have $J_m$ and $J_n$, the

$$S_g[m,n] = J_m \cdot J_n^{\top}, \tag{6}$$

where $\mathbf{S_g}$ is the ground truth similarity of $\mathbf{S_p}$.

## 3.4 | Loss functions

The loss functions we implement in this article can be divided into three categories, main training objective, auxiliary training objective, and contrastive training objective. The $\lambda$s noted in the following formulas are hyper-parameters to balance the losses.

$$L = L_{\text{main}} + L_{\text{auxiliary}} + L_{\text{DGCLNet}}. \tag{7}$$

### 3.4.1 | Main training objective

The main training objective contains three parts, 3D hand joints loss, 3D hand mesh vertices loss, and 2D hand joints loss. These three losses make sure the model performs a basic level of hand estimation. This part of the loss is calculated by minimizing the distance between the results of the main pipeline and the ground truth label.

$$L_{\text{main}} = \lambda_1 L_{\text{3D joints}} + \lambda_2 L_{\text{3D verts}} + \lambda_3 L_{\text{2D joints}}. \tag{8}$$

Note that the location of 3D hand joints and 3D hand mesh vertices is aligned with a root joint and is scaled by a length parameter. Therefore the two loss only describe the pose information of hand. The absolute position information and the scale information are learned by 2D hand joints loss. Besides, the camera parameter and rotation matrix are also learned by 2D hand joints loss, which can make projections to the 2D plane of images for the use of demonstration. We use L1 loss for all the three losses of the main training objective. In detail, it is L1 distance between estimated results and ground truth labels.

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### 3.4.2 | Auxiliary training objective

The auxiliary training objective contains three parts, length loss, consistent loss, and transition loss. These three parts improve the model performance in some aspects.

$$L_{\text{auxiliary}} = \lambda_4 L_{\text{length}} + \lambda_5 L_{\text{consistent}} + \lambda_6 L_{\text{transition}}, \tag{9}$$

$$L_{\text{consistent}} = \|J_k - \overline{J_k}\|_1, \tag{10}$$

The length loss here refers to the distance between joints. It can not only learn the scale of the hands but also can regularize the geometric structure of the hands. The consistent loss means the consistency between the graph features and the hand pose that the feature represents. With the help of the consistent loss, the model will be more stable and interpretable. The difference between the consistent loss and the 3D joints loss mentioned above is whether the GCN module is used. Consistent loss is calculated linearly from the feature generated by graph u-net while 3D joints loss is from MANO upsample layer. The former only contains pose information while the latter decodes pose information to precise 3D location. It can be calculated by the L1 distance between joint locations $J_k$ and ground truth labels $\overline{J}_k$. And the transition loss is the relative translation of each pair of vertices from left and right hand, which is a widely used restriction for two hand reconstruction mission.

### 3.4.3 | Contrastive training objective

$$L_{\text{DGCLNet}} = -\lambda_7 (L_{\text{cross_contrast}} + L_{\text{self_contrast}}), \tag{11}$$

$$L_{\text{cross_contrast}} = \|S_p - S_g\|_2, \tag{12}$$

$$L_{\text{self_contrast}} = log \frac{exp(S/\tau)}{\sum exp(S/\tau)}. \tag{13}$$

Note that $L_{\text{cross_contrast}}$ and $L_{\text{self_contrast}}$ describe cross-pose contrastive loss and self-pose contrastive loss. The cross-pose loss is evaluated by the similarity distinguish module, which contains $\mathbf{S_g}$ and $\mathbf{S_p}$. The self-pose contrastive loss is calculated by maximizing the similarity of input and output of the graph u-net. In order to utilize large enough batch size in our implementation, we adopt MoCo-like[34] dictionary settings to save the graph features of recent batches.

## 4 | EXPERIMENTS

## 4.1 | Experimental settings

### 4.1.1 | Implementation details

Our network is implemented using Pytorch. And the main model structure as well as the training pipeline is based on Pytorch lightning. All experiments are conducted on 4 NVIDIA RTX 2080ti GPUs. Training minibatch size is set as 32 and Adam is used for model optimization. The learning rate is initialized with $1 \times 10^{-4}$, and decays by half when the training loss decreases less than $10^{-4}$ continuously in 4 epoch. The minimum of learning rate is set by $10^{-6}$. The whole training takes 50 epochs, which takes about 2 days. We take input image of $256 \times 256$. The encoder is ResNet50 pretrained on ImageNet. And the GCN decoder is Chebyshev spectral graph CNN that is initialized randomly. $\lambda_1 = \lambda_2 = 10$ for 3D joints loss and 3D vertices loss. $\lambda_3 = \lambda_4 = 10^{-3}$ for 2D joints loss and length loss. $\lambda_5 = \lambda_6 = 1$ for consistent loss and transition loss. And $\lambda_7 = 10^{-2}$ for contrastive loss. The MoCo-like dictionary is implemented by a queue of 2048 and the model updates momentum.

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**TABLE 1** Comparison with state-of-the-art methods on InterHand2.6M.

| Methods | MPJPE↓ | MPVPE↓ |
|---|---|---|
| Boukhayma et al.[1] | 16.63 | 17.98 |
| Zhang et al.[8] | 13.48 | 13.95 |
| Rong et al.[9] | 13.56 | - |
| Kim et al.[44] | 12.08 | - |
| Meng et al.[45] | 10.97 | - |
| Li et al.[11] | **8.79** | **9.03** |
| Ours | **8.86** | **9.14** |

*Note*: The MPJPE and MPVPE are reported in mm. Bold values mean the top two greatest performance.

#### 4.1.2 | Evaluation metrics

As shown in Table 1, we used the mean per joint position error (MPJPE) to evaluate the hand pose estimation performance. MPJPE is defined as the mean Euclidean distance between ground truth and predicted 3D joint positions. Similarly the mean per vertex position error (MPVPE) is used to evaluate the hand shape reconstruction performance. Note that MPJPE and MPVPE in this article are reported after aligning the root joint and scaling the length of the middle metacarpal for fair comparison. All the measures mentioned above are calculate in millimeters.

### 4.2 | Dataset

Our model is trained and tested on InterHand2.6M[18] following the dataset settings of Reference 11. 366K training samples and 261K testing samples are picked out from the original InterHand2.6M dataset. And these samples are all the interacting two-hand data with annotation by both human and machine. The image samples are cropped and resize to $256 \times 256$ resolution .

### 4.3 | Qualitative results

Figure 3 shows our qualitative results for hand shape reconstruction. We compare the performance of our model to state-of-the-art reconstruction methods, which can be integrated with virtual reality (VR) or augmented reality (AR) applications. It is clear that our model offers a much better result in the details of processing double hands, especially the interactions among them, which makes the hand mesh closer to the natural state. Our model demonstrates the ability to successfully capture and reconstruct complex interactions of two hands. The first column of our qualitative results indicates that our model has fewer collisions. The results of Columns 2 and 4 indicate that the performance of our model is superior when the two hand meshes are close to each other, yet not making contact. Columns 3 and 5 indicate that our model results in a better matching between the hand mesh and the hand mask, which implies enhanced location accuracy. The last column of our results highlights that our model has a deep understanding of hand occlusion, and is able to reconstruct detailed hand poses competently.

### 4.4 | Ablation study

#### 4.4.1 | Baseline GCN

Before implementing our contrastive method, we run a GCN decoder as a baseline for following comparison. The result is shown in Table 2. The GCN is built on the base of References 2 and 11. Moreover, the graph Laplacian matrix obtained

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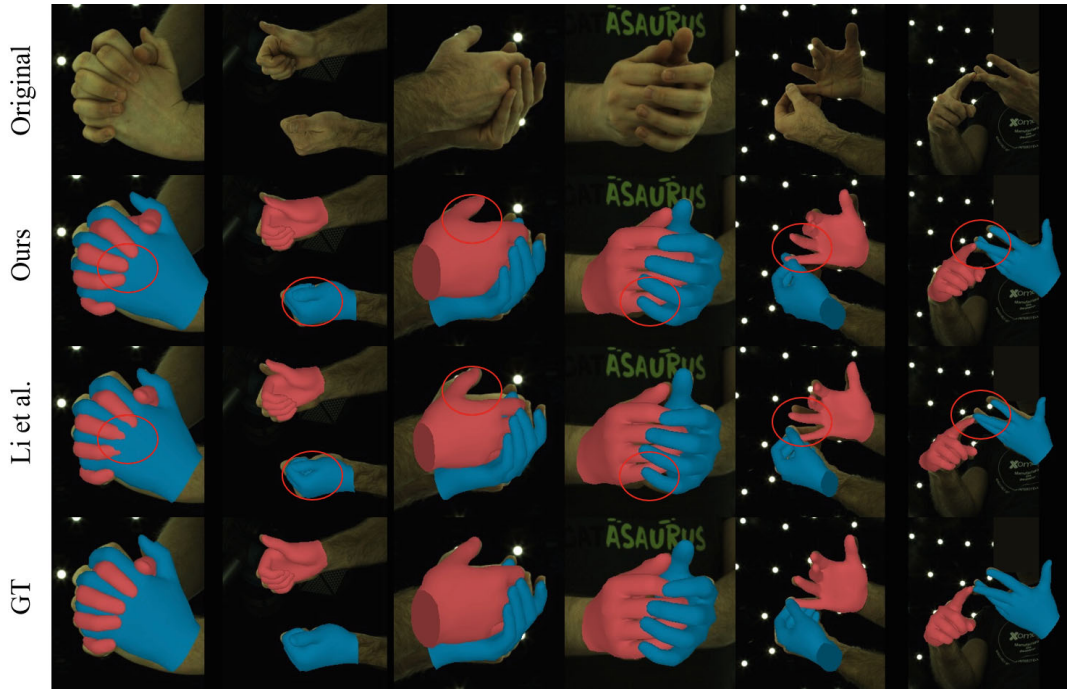



**FIGURE 3** Qualitative results of our method on InterHand2.6M test dataset and comparison with SOTA method proposed by Li et al.[11]

**TABLE 2** Ablation study of module choice on InterHand2.6M.

| | MPJPE↓ | MPVPE↓ |
|---|---|---|
| GCN baseline | 9.97 | 10.63 |
| GCN + random mask | 10.35 | 10.65 |
| GCN + random noise | 10.56 | 10.89 |
| GCN + random edge | 10.59 | 10.93 |
| GCN + flip edge | 10.92 | 11.30 |
| GCN + DGCLNet | **9.02** | **9.31** |
| GCN + DGCLNet (ours)[a] | **8.86** | **9.14** |

*Note*: Bold values mean the top two greatest performance.
[a]Similarity distinguish strategy is used.

from MANO-pretraining was fixed during training. We observe that the baseline is strong, but may struggle with fingertip alignment and encountering obstacles.

### 4.4.2 | Adding contrastive module

We tested the contrastive module with both EGD and DGCLNet based on GCN baseline. The result is shown in Table 2. As discussed in Section 3.4, explicit graph disturbing methods may lead to a loss of consistency for graph features which accounts for their poor performance. We see that DGCLNet module reduces the error by 1.6 mm. Thanks to the disturbance network, the model is able to learn from slightly different graph features without compromising the consistency of the pose and feature. The qualitative results are given in Figure 4. It is clear that our DGCLNet module improves the baseline by reducing collisions, resulting in a hand mesh that is more realistic and lifelike. Besides, the overlap of hand mask is more precise, which indicates the higher performance in visual comprehension.

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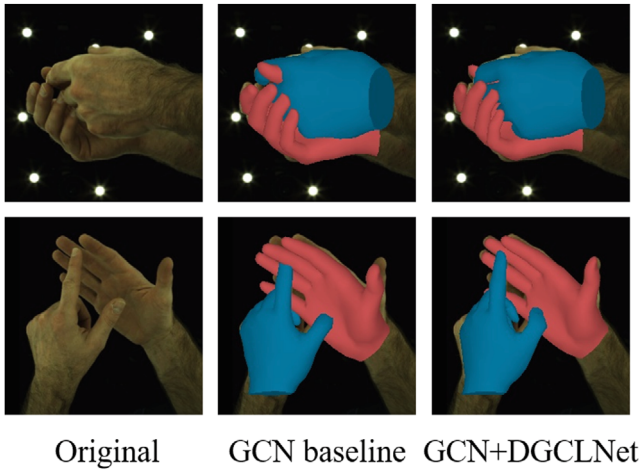



**FIGURE 4** Qualitative ablation study on InterHand2.6M.

### 4.4.3 | Similarity distinguish

Our similarity distinguish method makes the target of contrastive pairs more reliable and yields better robustness while testing. Contrastive learning can be notoriously difficult to converge and can easily become trapped in severe oscillations. We manage to solve this problem by making the similarity of the pose of embedded feature the target similarity of embedded graph feature. The ablation result is shown in Table 2.

## 5 | CONCLUSION

In conclusion, this article introduces a novel graph contrastive method for two-hands 3D reconstruction from a single image. To deal with challenges such as hand-hand occlusion and the homogeneous and self-similar appearance of hands, we introduce novel graph contrastive method to enhance model ability to rebuild 3D topographical structure. More specifically, an explicit graph disturbing is tried but does not perform well due to its inconsistency. To resolve that issue, a DGCLNet was proposed to force the features of disturbed data to remain consistent. A similarity distinguish strategy was further used instead of the discrete settings which made the model more robust and yielded better results. Moreover, a multi-term loss is designed to balance the relation among the hand pose, the visual scale and the viewpoint position. Experiments validate the superiority of our proposed method over the state-of-the-art methods.

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## DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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## REFERENCES

1. Boukhayma A, de Bem R, Torr PHS. 3D hand shape and pose from images in the wild. Proceedings of the 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR); 2019. p. 10835–44.

1546427x, 2023, 3-4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cav.2186 by University Of Science & Tech Beijin, Wiley Online Library on [12/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

2. Ge L, Ren Z, Li Y, Xue Z, Wang Y, Cai J, et al. 3D hand shape and pose estimation from a single RGB image. Proceedings of the 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR); 2019. p. 10825–34.
3. Baek S, In Kim K, Kim T-K. Pushing the envelope for RGB-based dense 3D hand pose estimation via neural rendering. Proceedings of the 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR); 2019. p. 1067–76.
4. Baek S, In Kim K, Kim T-K. Weakly-supervised domain adaptation via GAN and mesh model for estimating 3D hand poses interacting objects. Proceedings of the 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR); 2020. p. 6120–30.
5. Wan C, Probst T, Van Gool L, Yao A. Dual grid net: hand mesh vertex regression from single depth maps. In: Vedaldi A, Bischof H, Brox T, Frahm JM, editors. Computer Vision – ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XXX. Berlin: Springer-Verlag; 2020. p. 442–59.
6. Zhou Y, Habermann M, Xu W, Habibie I, Theobalt C, Feng X. Monocular real-time hand shape and motion capture using multi-modal data. Proceedings of the 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR); 2020. p. 5345–54.
7. Karunratanakul K, Spurr A, Fan Z, Hilliges O, Tang S. A skeleton-driven neural occupancy representation for articulated hands. Proceedings of the 2021 International Conference on 3D Vision (3DV); 2021. p. 11–21.
8. Zhang B, Wang Y, Deng X, Zhang Y, Tan P, Ma C, et al. Interacting two-hand 3D pose and shape reconstruction from single color image. Proceedings of the 2021 IEEE/CVF International Conference on Computer Vision (ICCV); 2021. p. 11334–43.
9. Rong Y, Wang J, Liu Z, Loy CC. Monocular 3D reconstruction of interacting hands via collision-aware factorized refinements. Proceedings of the 2021 International Conference on 3D Vision (3DV); 2021. p. 432–41.
10. Romero J, Tzionas D, Black MJ. Embodied hands: modeling and capturing hands and bodies together. ACM Trans Graph. 2017; 36(6):245.
11. Li M, Liang A, Zhang H, Wu L, Chen F, Yu T, et al. Interacting attention graph for single image two-hand reconstruction. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR); June 2022. p. 2761–70.
12. Sridhar S, Mueller F, Zollhöfer M, Casas D, Oulasvirta A, Theobalt C. Real-time joint tracking of a hand manipulating an object from RGB-D input. In: Leibe B, Matas J, Sebe N, Welling M, editors. Computer Vision – ECCV 2016. Cham: Springer International Publishing; 2016. p. 294–310.
13. Zhang J, Jiao J, Chen M, Liangqiong Q, Xu X, Yang Q. A hand pose tracking benchmark from stereo matching. Proceedings of the 2017 IEEE International Conference on Image Processing (ICIP); 2017. p. 982–6.
14. Mueller F, Mehta D, Sotnychenko O, Sridhar S, Casas D, Theobalt C. Real-time hand tracking under occlusion from an egocentric RGB-D sensor. Proceedings of the 2017 IEEE International Conference on Computer Vision (ICCV); 2017. p. 1163–72.
15. Zimmermann C, Brox T. Learning to estimate 3D hand pose from single RGB images. Proceedings of the 2017 IEEE International Conference on Computer Vision (ICCV); 2017. p. 4913–21.
16. Joo H, Simon T, Li X, Liu H, Tan L, Gui L, et al. Panoptic studio: a massively multiview system for social interaction capture. IEEE Trans Pattern Anal Mach Intell. 2019;41(1):190–204.
17. Zimmermann C, Ceylan D, Yang J, Russell B, Argus MJ, Brox T. FreiHAND: a dataset for markerless capture of hand pose and shape from single RGB images. Proceedings of the 2019 IEEE/CVF International Conference on Computer Vision (ICCV); 2019. p. 813–22.
18. Moon G, Yu S-I, He W, Shiratori T, Lee KM. InterHand2.6M: a dataset and baseline for 3D interacting hand pose estimation from a single RGB image. Proceedings of the European Conference on Computer Vision (ECCV); 2020.
19. Sinha A, Choi C, Ramani K. DeepHand: robust hand pose estimation by completing a matrix imputed with deep features. Proceedings of the 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR); 2016. p. 4150–8.
20. Ge L, Liang H, Yuan J, Thalmann D. Robust 3D hand pose estimation in single depth images: from single-view CNN to multi-view CNNs. Proceedings of the 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR); 2016. p. 3593–601.
21. Malik J, Elhayek A, Nunnari F, Varanasi K, Tamaddon K, Heloir A, et al. DeepHPS: end-to-end estimation of 3D hand pose and shape by learning from synthetic depth. Proceedings of the 2018 International Conference on 3D Vision (3DV); 2018. p. 110–19.
22. Spurr A, Song J, Park S, Hilliges O. Cross-modal deep variational hand pose estimation. Proceedings of the 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition; 2018. p. 89–98.
23. Simon T, Joo H, Matthews I, Sheikh Y. Hand keypoint detection in single images using multiview bootstrapping. Proceedings of the 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR); 2017. p. 4645–53.
24. Iqbal U, Molchanov P, Breuel T, Gall J, Kautz J. Hand pose estimation via latent 2.5D heatmap regression. In: Ferrari V, Hebert M, Sminchisescu C, Weiss Y, editors. Computer Vision – ECCV 2018. Cham: Springer International Publishing; 2018. p. 125–43.
25. Wang J, Mueller F, Bernard F, Sorli S, Sotnychenko O, Qian N, et al. RGB2Hands: real-time tracking of 3D hand interactions from monocular RGB video. Vol 39. New York: Association for Computing Machinery; 2020.
26. He Y, Wei H. 3D hand pose estimation via regularized graph representation learning. In: Fang L, Chen Y, Zhai G, Wang J, Wang R, Dong W, editors. Artificial intelligence. Cham: Springer International Publishing; 2021. p. 540–52.
27. Hampali S, Sarkar SD, Rad M, Lepetit V. Keypoint transformer: solving joint identification in challenging hands and object interactions for accurate 3D pose estimation. Proceedings of the IEEE Computer Vision and Pattern Recognition Conference, 2022. p. 11090-100.
28. Spurr A, Dahiya A, Wang X, Zhang X, Hilliges O. Self-supervised 3D hand pose estimation from monocular RGB via contrastive learning. Proceedings of the 2021 IEEE/CVF International Conference on Computer Vision (ICCV); 2021. p. 11210–19.
29. Defferrard M, Bresson X, Vandergheynst P. Convolutional neural networks on graphs with fast localized spectral filtering. Proceedings of the 30th International Conference on Neural Information Processing Systems. Red Hook, NY: Curran Associates Inc. 2016. p. 3844–52.

1546427x, 2023, 3-4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cav.2186 by University Of Science & Tech Beijin, Wiley Online Library on [12/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

30. Kulon D, Wang H, Güler RA, Bronstein MM, Zafeiriou S. Single image 3D hand reconstruction with mesh convolutions. Proceedings of the British Machine Vision Conference (BMVC); 2019.
31. Park JK, Yeonguk O, Moon G, Choi H, Lee KM. HandOccNet: occlusion-robust 3D hand mesh estimation network. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR); June 2022. p. 1496–505.
32. Chen T, Kornblith S, Norouzi M, Hinton G. A simple framework for contrastive learning of visual representations. Proceedings of the 37th International Conference on Machine Learning; 2020.
33. Chen T, Kornblith S, Swersky K, Norouzi M, Hinton GE. Big self-supervised models are strong semi-supervised learners. In: Larochelle H, Ranzato M, Hadsell R, Balcan MF, Lin H, editors. Advances in neural information processing systems. Volume 33. Red Hook, NY: Curran Associates, Inc.; 2020. p. 22243–55.
34. He K, Fan H, Wu Y, Xie S, Girshick R. Momentum contrast for unsupervised visual representation learning. Proceedings of the 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR); 2020. p. 9726–35.
35. Grill J-B, Strub F, Altché F, Tallec C, Richemond P, Buchatskaya E, et al. Bootstrap your own latent – a new approach to self-supervised learning. In: Larochelle H, Ranzato M, Hadsell R, Balcan MF, Lin H, editors. Advances in neural information processing systems. Volume 33. Red Hook, NY: Curran Associates, Inc.; 2020. p. 21271–84.
36. Zimmermann C, Argus M, Brox T. Contrastive representation learning for hand shape estimation. In: Bauckhage C, Gall J, Schwing A, editors. Pattern recognition. Cham: Springer International Publishing; 2021. p. 250–64.
37. You Y, Chen T, Sui Y, Chen T, Wang Z, Shen Y. Graph contrastive learning with augmentations. In: Larochelle H, Ranzato M, Hadsell R, Balcan MF, Lin H, editors. Advances in neural information processing systems. Volume 33. Red Hook, NY: Curran Associates, Inc.; 2020. p. 5812–23.
38. Chu G, Wang X, Shi C, Jiang X. CuCo: graph representation with curriculum contrastive learning. In: Zhou Z-H, editor. Proceedings of the Thirtieth International Joint Conference on Artificial Intelligence; 2021. p. 2300–6.
39. Thakoor S, Tallec C, Azar MG, Azabou M, Dyer EL, Munos R, et al. Large-scale representation learning on graphs via bootstrapping. Proceedings of the International Conference on Learning Representations; 2022.
40. Lee N, Lee J, Park C. Augmentation-free self-supervised learning on graphs. Proceedings of the Thirty-Sixth AAAI Conference on Artificial Intelligence (AAAI-22); 2022. p. 7372–80.
41. Tong Z, Liang Y, Ding H, Dai Y, Li X, Wang C. Directed graph contrastive learning. In: Ranzato M, Beygelzimer A, Dauphin Y, Liang PS, Wortman Vaughan J, editors. Advances in neural information processing systems. Volume 34. Red Hook, NY: Curran Associates, Inc.; 2021. p. 19580–93.
42. Chen X, He K. Exploring simple Siamese representation learning. Proceedings of the 2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR); 2021. p. 15745–53.
43. Gao H, Ji S. Graph u-nets. Proceedings of the International Conference on Machine Learning; 2019. p. 2083–92.
44. Kim DU, In Kim K, Baek S. End-to-end detection and pose estimation of two interacting hands. Proceedings of the 2021 IEEE/CVF International Conference on Computer Vision (ICCV); 2021. p. 11169–78.
45. Meng H, Jin S, Liu W, Qian C, Lin M, Ouyang W, et al. 3D interacting hand pose estimation by hand de-occlusion and removal. In: Avidan S, Brostow G, Cissé M, Farinella GM, Hassner T, editors. Computer vision – ECCV 2022. Berlin: Springer-Verlag; 2022. p. 380–97.

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